

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of M. Sc. (AOC) Examination November 2019

OC 822: Natural Products

Date: 08/11/2019 Day: Friday Time: 01.30 p.m. to 02.00 p.m. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q-I Draw the structure of following compounds. 20

1 Methyl morphenol

2 Apomorphine

3 Quinine

4 β -eudesmol

5 Meroquinene

6 Cholesterol

7 Lithocholic acid

8 Estrone

9 β -ionone

10 Cinchonine

11 Cholanic acid

12 Thebaine

13 Testosterone

14 Adenine

15 Codeine

16 Coprostane

17 Guanine

18 Menthol

19 Androsterone

20 Morphine

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III Semester of M. Sc. (AOC) Examination November 2019

OC 822: Natural Products

Date: 08/11/2019 Day: Friday Time: 02.00 p.m. to 04.30 p.m. Maximum Marks: 50

Important Instructions:

- Section I and II must be attempted in TWO ANSWER SHEET.
 - Make suitable assumptions and draw neat figures wherever required.
 - Use of non-programmable calculator is allowed.
 - Show necessary calculations.
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SECTION-I

Q-II Answer the following questions as directed.

- A Discuss the relationship between morphine, codeine and thebaine. Show the important chemical transformations of these alkaloids used in structure determination. 04
- B Give the mechanism for the formation of apomorphine from morphine under acid treatment. 03
- C Explain: Hofmann exhaustive methylation is an important tool in the study of the structure of alkaloid. 03

OR

- C Explain: von Braun method for structure determination of alkaloid. 03

Q-III Answer the following questions as directed.

- A Discuss the structure of meroquinene in detail. 04
- B Give the mechanism for the formation of thebanine from thebaine upon treatment of acid. 03
- C Discuss the presence of different functional groups in quinine. 03

OR

- C Discuss the presence of different functional groups in cinchonine. 03

SECTION-II

Q-IV Answer the following questions as directed.

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| A | Discuss the position of double bond in β -eudesmol. | 04 |
| B | Give the classification of terpenoids. | 03 |
| C | Write a short note on isoprene rule. | 03 |

OR

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|----------|----------------------------------|-----------|
| C | Write the synthesis of eudalene. | 03 |
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Q-V Answer the following questions as directed.

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|----------|---|-----------|
| A | Discuss the position of 2°-OH and double bond in cholesterol. | 04 |
| B | Discuss the nature of side chain in cholesterol. | 03 |
| C | Discuss the relationship between bile acids and sterols. | 03 |

OR

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| C | Describe the chemical relationship between estrone, estriol and estradiol. | 03 |
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Q-VI Answer the following questions as directed.

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| A | Explain the working mechanism of enzyme and list out the factors affecting rate of enzyme catalyzed reaction. | 04 |
| B | Write a short note on competitive and noncompetitive inhibitors. | 03 |
| C | Discuss the biogenesis of terpenoids through mevalonic acid pathway. | 03 |

OR

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| C | Discuss the biogenesis of cholesterol. | 03 |
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